In [36]:
```
#1 Read a file containing student names (one per line) and storethem in a list.

with open("demo.txt","w") as f:
    f.write("John")
    f.write("\nGus")
    f.write("\nPablo")
    f.write("\nGus")
    f.write("\nnitin")
    f.write("\nAemond")
    f.write("\nDaemon")

with open("demo.txt") as f:
    std_name = f.readlines()

print("Student name as a list :")
for std in std_name:
    print(std,end=" ")
```

```
Student name as a list :
John
 Gus
 Pablo
 Gus
 nitin
 Aemond
 Daemon
```

In [37]:
```
#2 Print names longer than 5 characters.

with open("demo.txt") as f:
    std_name = f.readlines()

print ("Names Longer than 5 Chars :")
for std in std_name:
    if len(std) > 5:
        print(std,end=" ")
```

```
Names Longer than 5 Chars :
Pablo
 nitin
 Aemond
 Daemon
```

In [35]:
```
#3 Count how many names start with a vowel

with open("demo.txt") as f:
    std_name = f.readlines()
# print(std_name)
vowels = "aeiou"
print("Students with Vowel as Starting Char in Names :")
for student in std_name:
    if student[0] in vowels or student[0] in vowels.upper():
        print(student,end= "")
```

```
Students with Vowel as Starting Char in Names :
Aemond
```

In [11]:
```
#4 Save the filtered names (longer than 5 characters) into a newfile.
```

```
with open("demo.txt") as f:
    std_name = f.readlines()
filtered_name=[]
for student in std_name:
    if len(student)>5:
        filtered_name.append(student)

with open("filtered_name.txt","w") as f:
    f.writelines(filtered_name)

print("Filter Name as Content :")
with open("filtered_name.txt")as f:
    print(f.read())
```

```
Filter Name as Content :
Pablo
Aemond
Daemon
```

In [16]:
```
#5 Sort the list of names alphabetically and print them in ascending order.

with open("demo.txt") as f:
    std_name = f.readlines()
sorted_name = sorted(std_name)
for student in sorted_name:
    print(student,end=" ")
```

```
Aemond
Daemon Gus
John
Pablo
```

In [13]:
```
#6 Print names that contain the letter "a".

with open("demo.txt") as f:
    std_name = f.readlines()
for student in std_name:
    if 'a' in student:
        print (student,end=" ")
```

```
Pablo
Daemon
```

In [14]:
```
#7 Count word frequency in names (handle duplicates) and store results in a dict

with open("demo.txt") as f:
    std_name = f.read()
processed_name = std_name.replace("\n"," ")
mapping = {}
for student in processed_name.split():
    if student.strip() not in mapping:
        mapping[student.strip()] = 1
    else:
        mapping[student.strip()] += 1
print("Word frequency in dict with names & count: ")
for key,value in mapping.items():
    print(f"{key}:{value}")
```

```
Word frequency in dict with names & count:
John:1
Gus:1
Pablo:1
Aemond:1
Daemon:1
```

In [38]:
```python
#8 String Analysis -
# For each name, print:
#   length of the name
#   First and last character
#   Whether it is a palindrome

def is_palindrome(strs):
    for i in range(len(strs)):
        if name[i] != name[len(name)-i -1]:
            return False
    return True
with open("demo.txt") as f:
    std_name = f.read()
processed_name = std_name.replace("\n"," ")
print("Iteration over each Name :")
for name in processed_name.split():
    print("-"*12)
    print("name :",name)
    print("Length of the Name :",len(name))
    print("First Character :",name[0])
    print("Last Character :",name[-1])

    print(f"is {name} a is_palindrome ? {is_palindrome(name)}")
```

```
Iteration over each Name :
------------
name : John
Length of the Name : 4
First Character : J
Last Character : n
is John a is_palindrome ? False
------------
name : Gus
Length of the Name : 3
First Character : G
Last Character : s
is Gus a is_palindrome ? False
------------
name : Pablo
Length of the Name : 5
First Character : P
Last Character : o
is Pablo a is_palindrome ? False
------------
name : Gus
Length of the Name : 3
First Character : G
Last Character : s
is Gus a is_palindrome ? False
------------
name : nitin
Length of the Name : 5
First Character : n
Last Character : n
is nitin a is_palindrome ? True
------------
name : Aemond
Length of the Name : 6
First Character : A
Last Character : d
is Aemond a is_palindrome ? False
------------
name : Daemon
Length of the Name : 6
First Character : D
Last Character : n
is Daemon a is_palindrome ? False
```

In [30]:

```python
#9 File + List + String Combined Challenge
#   Read names from file → list.
#   Remove duplicates.
#   Sort by length (shortest to longest).
#   Write results into processed_names.txt

with open("demo.txt") as f:
    std_name = f.readlines()
processed_name = [name.strip() for name in std_name]
processed_name = list(set(processed_name))
processed_name.sort(key=len)

with open ("processed_name.txt","w") as f:
    for name in processed_name:
        f.write(name+"\n")
    print(processed_name)
```

```
['Gus', 'John', 'Pablo', 'Daemon', 'Aemond']
```

In [43]:
```python
#10 Read names from file.
#     Group them by starting letter (e.g., all names starting with A together).
#     Save each group into separate files (A_names.txt, B_names.txt, etc.).

with open("demo.txt") as f:
    std_name = f.readlines()
processed_name = [name.strip()for name in std_name]
group = {}
for name in processed_name:
    letter = name[0].upper()
    if letter not in group:
        group[letter] = []
        group[letter].append(name)
        for key,  value in group.items():
            with open(key + "_names.txt","w") as f:
                for name in value:
                    f.write(name + "\n")
print("Files Created !!")
```

```
Files Created !!
```